

Early prediction of in-hospital mortality using routinely collected electronic health record data: an analysis of the MIMIC-III database

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Abstract

Early prediction of mortality in critically ill patients remains a major challenge in intensive care medicine. Although commonly used severity scoring systems provide standardized risk estimates, their reliance on fixed assumptions may limit adaptability across patient populations and clinical settings. The growing availability of electronic health record data offers an opportunity to develop data-driven models that leverage routinely collected clinical information for early risk stratification.

In this study, we used the MIMIC-III database to develop an interpretable statistical model for predicting in-hospital mortality based on demographic characteristics, vital signs, and laboratory measurements obtained within the first 24 hours of ICU admission. By emphasizing transparency and clinical interpretability, this approach aims to complement existing risk assessment tools and support early clinical decision-making in critically ill patients.

Non- Scientific Abstract

Patients admitted to the intensive care unit (ICU) are often severely ill, and predicting who is at higher risk of dying remains a major challenge for healthcare teams. Being able to identify high-risk patients early may help clinicians make better decisions, prioritize resources, and intervene sooner to prevent clinical deterioration. Although several scoring systems are commonly used to estimate mortality risk in the ICU, they are based on fixed rules and may not perform equally well for all patient groups or changing clinical settings.

With the increasing use of electronic health records, large amounts of patient data are now available for analysis. These data make it possible to develop prediction models that are designed for specific populations using clinical information. In this study, we used data from a large publicly available ICU database to explore whether early physiological and laboratory measurements can help predict in-hospital mortality.

We developed a statistical prediction model using information collected during the first 24 hours of ICU admission, such as vital signs and laboratory results. The model was evaluated based on its ability to distinguish between patients who survived and those who did not. By focusing on a transparent and interpretable approach, this work aims to show how simple, data-driven models can

support early risk assessment in critically ill patients and potentially assist clinicians in identifying those who may require closer monitoring or more aggressive treatment.

Introduction

Mortality prediction in the intensive care unit (ICU) remains a central challenge in critical care medicine. Early identification of patients at high risk of death may support clinical decision-making, guide allocation of healthcare resources, and enable timely interventions aimed at preventing clinical deterioration [1,2]. Despite advances in monitoring and treatment, in-hospital remains substantial, particularly among patients presenting with severe physiological derangements and multi-organ dysfunction [3].

Several prognostic scoring systems have been developed to estimate mortality risk in critically ill patients. Among the most widely used are the Acute Physiology and Chronic Health Evaluation (APACHE), the Simplified Acute Physiology Score (SAPS II), and the Sequential Organ Failure Assessment (SOFA) score [4–6]. These tools rely on physiological and laboratory variables collected during the first hours of ICU admission and have demonstrated good predictive performance across diverse patient populations. However, their fixed structure and limited adaptability may restrict their performance in specific cohorts or evolving clinical contexts [7].

The growing availability of electronic health record (EHR) data has enabled the application of data-driven approaches to outcome prediction in critical care. Large clinical databases allow the analysis of high-dimensional, longitudinal patient data and facilitate the development of predictive models tailored to specific populations [8]. In this context, the Medical Information Mart for Intensive Care III (MIMIC-III) database represents a landmark resource, providing de-identified clinical data from over 40,000 ICU patients and supporting reproducible critical care research [9].

While machine learning methods have gained popularity for mortality prediction, traditional statistical approaches such as logistic regression remain highly relevant in clinical research. Logistic regression offers robustness, transparency, and interpretability, allowing direct estimation of the independent association between predictors and outcomes [10]. In healthcare settings, where explainability is essential for clinical trust and adoption, interpretable models may offer advantages over more complex black-box algorithms [11].

The aim of this study is to develop and evaluate a predictive model for in-hospital mortality using routinely collected physiological and laboratory variables from the first 24 hours of ICU admission, extracted from the MIMIC-III database. By combining clinical insight with a data-driven analytical framework, we seek to identify key early predictors of mortality and assess the discriminatory performance of a multivariate logistic regression model. This work illustrates the potential of EHR-based analyses to support early risk stratification in critical care while maintaining clinical interpretability.

Methods

Data source

This retrospective study was conducted using data extracted from the Medical Information Mart for Intensive Care III (MIMIC-III) database, a large, publicly available repository containing de-identified clinical data from patients admitted to intensive care units at the Beth Israel Deaconess Medical Center between 2001 and 2012 [9]. The database includes detailed information on demographics, vital signs, laboratory measurements, diagnoses, procedures, and outcomes, and has been extensively used in critical care and electronic health record (EHR) research.

Access to the MIMIC-III database was obtained after completion of the required data use agreement and ethics training. As all data are de-identified, institutional review board approval and informed consent were not required, in accordance with established guidelines [9].

The dataset used in this study was obtained through structured SQL queries, fully documented in Supplementary Notebook S1.

Study population

The study population consisted of adult patients admitted to the intensive care unit. To ensure independence of observations and avoid intra-patient correlation, only the first ICU admission for each patient was included in the analysis. Patients with missing mortality outcome data were excluded.

A retrospective cohort design was employed. Clinical data recorded during the first 24 hours of ICU admission were analyzed in order to capture early physiological status and align with established ICU risk stratification practices [4–6].

Predictor variables and outcome

The primary outcome of the study was in-hospital mortality (*hospital_expire_flag*). Predictor variables were selected based on clinical relevance and availability within the first 24 hours of ICU admission. These included demographic characteristics (age, gender), vital signs (heart rate, respiratory rate, mean arterial pressure, temperature, SpO₂) and laboratory markers (lactate, creatinine, white blood cell count).

When multiple measurements of the same variable were available during the first 24 hours, mean values computed during the SQL extraction phase were used to summarize early physiological status. Continuous variables were retained in their original scale to preserve clinical interpretability. Variable selection aimed to balance clinical plausibility, data completeness, and consistency with variables commonly used in established ICU severity scoring systems [4–6].

Data Preprocessing and Outlier Handling

Data quality is a critical challenge in EHR analysis. In accordance with MIMIC-III privacy guidelines, ages originally coded as > 89 years were adjusted to 91 years. To manage extreme physiological

values, which often represent critical illness rather than measurement error, without compromising statistical stability, we applied a Winsorization technique. Continuous variables were capped at the 1st and 99th percentiles. This approach replaces extreme outliers with the nearest boundary values, preserving the signal of physiological derangement while mitigating the influence of implausible artifacts. This method is preferred over removing outliers, as exclusion could bias the model by systematically removing the most critically ill patients.

Statistical analysis

Descriptive statistics were used to summarize the baseline characteristics. Continuous variables are reported as means with standard deviations (SD) and categorical variables as counts and percentages. A Pearson correlation matrix was computed to screen for multicollinearity; all predictor variables showed low correlation ($r < 0.35$), ensuring their independence in the multivariate model.

The dataset presented a significant class imbalance, with a mortality rate of 11%. To prevent the model from biasing predictions toward the majority class (survivors), we implemented a Weighted Logistic Regression model. Observations were assigned weights inversely proportional to their class frequency, imposing a higher penalty for misclassifying non-survivors. This strategy was chosen in order to prioritize sensitivity and early identification of high risk patients.

Model Evaluation and Reproducibility

The dataset was randomly divided into a training set (70%, $n = 22,401$) and a testing set (30%, $n = 9,599$). The distribution of the primary outcome was consistent across both subsets, with an in-hospital mortality rate of 10.96% ($n = 2,455$ in training; $n = 1,052$ in testing). Furthermore, the performance of the model was quantified using the Area Under the ROC Curve (AUC), which measures the ability of the model to discriminate between survivors and non-survivors [12]. Additionally, a confusion matrix was used to evaluate Sensitivity, Specificity and Negative Predictive Value (NPV).

Finally, all analyses were performed using the R statistical programming language. The analysis pipeline was implemented using open-source packages to ensure reproducibility and transparency. The full preprocessing workflow, modelling and evaluation pipeline in Supplementary Notebook S2. Together with Supplementary Notebook S1 (data extraction), these materials allow full reproduction of the study.

Results

Study population and Baseline Characteristics

A total of 32,000 ICU admissions were included (first ICU stay per patient). Overall, 28,493 (89.4%) survived and 3,507 (10.96%) patients died during the hospital stay (`hospital_expire_flag = 1`). As mentioned in before, the model was validated on an independent test set of 9,599 patients containing 1,052 deaths. Baseline demographic and early clinical variables are reported in Table 1. Compared with survivors, non-survivors were older (70.22 ± 15.77 vs 63.33 ± 17.29 years, $p < 0.001$) and had a higher proportion of females (47.1% vs 42.7%, $p < 0.001$). During the first 24 hours of ICU

admission, non-survivors showed higher heart rate (89.94 ± 17.28 vs 85.13 ± 14.74 bpm, $p < 0.001$) and lower mean arterial pressure (77.11 ± 10.84 vs 78.60 ± 9.30 mmHg, $p < 0.001$). Notably, non-survivors had higher lactate (2.63 ± 1.84 vs 1.96 ± 0.82 mmol/L, $p < 0.001$) and higher creatinine (1.63 ± 1.32 vs 1.20 ± 1.09 mg/dL, $p < 0.001$) levels than survivors.

Respiratory rate and SpO₂ showed unusually large standard deviations in survivors, suggesting the presence of outliers or data quality issues typical of retrospective EHR data.

Table 1. Baseline Characteristics and Clinical Variables Stratified by Hospital Mortality

Variable	Survivors (n=28,493)	Non-survivors (n=3,507)	p-value
Age (years), mean (SD)	63.33 (17.29)	70.22 (15.77)	< 0.001
Gender, n (%)			< 0.001
Female	12,175 (42.7%)	1,652 (47.1%)	
Male	16,318 (57.3%)	1,855 (52.9%)	
Heart Rate (bpm), mean (SD)	85.13 (14.74)	89.94 (17.28)	< 0.001
MAP (mmHg), mean (SD)	78.60 (9.30)	77.11 (10.84)	< 0.001
Respiratory Rate (bpm), mean (SD)	18.43 (3.70)	20.45 (4.60)	< 0.001
SpO ₂ (%), mean (SD)	97.41 (1.72)	96.95 (2.27)	< 0.001
Temperature (°F), mean (SD)	98.32 (1.03)	98.18 (1.41)	< 0.001
Lactate (mmol/L), mean (SD)	1.96 (0.82)	2.63 (1.84)	< 0.001
Creatinine (mg/dL), mean (SD)	1.20 (1.09)	1.63 (1.32)	< 0.001
WBC Count (10 ⁹ /L), mean (SD)	7.00 (0.28)	7.02 (0.63)	6

Note: SD: Standard Deviation; MAP: Mean Arterial Pressure; bpm: beats/ breaths per minute; WBC: White Blood Cell Count.

Multivariate logistic regression analysis

A multivariate weighted logistic regression was performed to identify independent predictors of in-hospital mortality using clinical variables recorded during the first 24 hours of ICU admission.

Results including coefficients (β), Odds Ratios (OR), and 95% Confidence Intervals (CI) are summarized in Table 2, with the corresponding association between variables and mortality illustrated in the Forest Plot (Figure 1).

In the multivariable logistic regression analysis, lactate emerged as the strongest independent predictor of mortality (OR 1.46, 95% CI 1.42–1.50), followed by creatinine (OR 1.31, 95% CI 1.27–1.34). Both variables showed robust associations, with confidence intervals entirely above the null value.

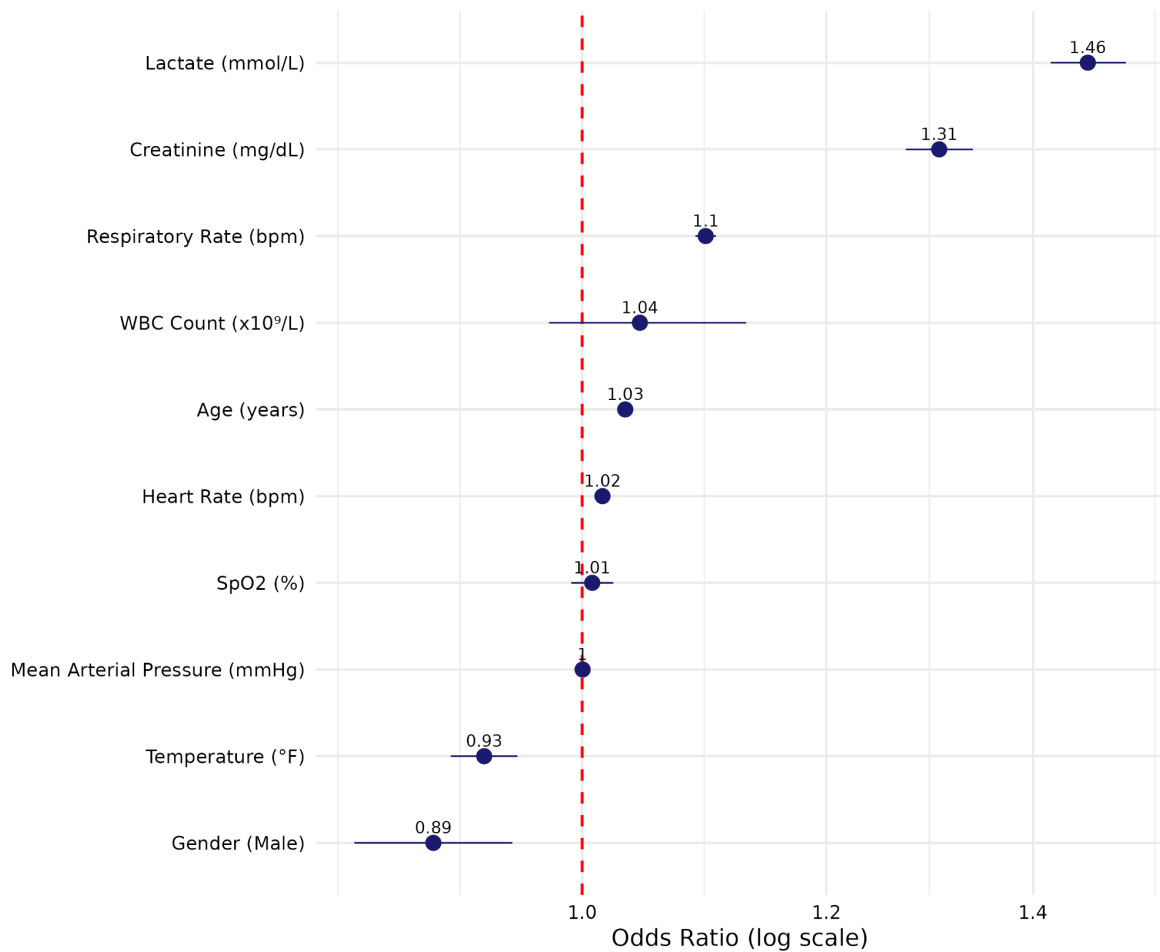
Among demographic and physiological parameters, age, heart rate, and respiratory rate were significantly associated with increased mortality risk, while male sex (OR 0.90, 95% CI 0.84–0.95; $p < 0.001$) and body temperature (OR 0.93, 95% CI 0.91–0.95) demonstrated a protective effect, being independently associated with lower odds of death.

In contrast, white blood cell count, mean arterial pressure, and oxygen saturation (SpO_2) were not significantly associated with mortality, as their 95% confidence intervals crossed the null value, indicating the absence of an independent effect in the adjusted model. The relative magnitude and direction of all associations are summarized in the corresponding figure.

Table 2. Multivariate Weighted Logistic Regression Results

Variable	Estimate (β)	Std. Error	p-value	Odds Ratio (95% CI)
(Intercept)	-2.860	1.4355	842	0.75 (0.05 – 12.52)
Age	321	10	< 0.001	1.03 (1.03 – 1.04)
Gender (Male)	-1.111	301	< 0.001	0.90 (0.84 – 0.95)
Heart Rate	152	10	< 0.001	1.02 (1.01 – 1.02)
Respiratory Rate	922	39	< 0.001	1.10 (1.09 – 1.11)
Temperature	-731	127	< 0.001	0.93 (0.91 – 0.95)
Lactate	3.775	143	< 0.001	1.46 (1.42 – 1.50)
Creatinine	2.665	128	< 0.001	1.31 (1.27 – 1.34)
WBC Count	431	366	239	1.04 (0.98 – 1.13)
MAP	3	15	835	1.00 (1.00 – 1.00)
SpO2	75	80	349	1.01 (0.99 – 1.02)

Figure 1. Forest plot showing adjusted odd ratios and 95% confidence intervals for early clinical predictors of in-hospital mortality.



Note: Estimates and 95% confidence intervals (CI) for clinical predictors recorded during the first 24h of ICU admission. The dashed vertical line at OR = 1 represents the null effect. Estimates positioned the right of the line indicate an increased risk of mortality, while those to the left indicate a protective effect. Variables whose 95% CI intersect the null line are not considered statistically significant.

Model performance and Evaluation

The discriminatory performance of the multivariate logistic regression model was assessed using receiver operating characteristic (ROC) curve analysis. The model's discriminatory power reached an Area Under the ROC curve (AUC) of 0.7255 (Figure 3.), indicating moderate-to-good discrimination between hospital survivors and non-survivors.

The classification results on the test set (n = 9,599) are detailed in Table 3 and illustrated in the confusion matrix heatmap (Figure 2.). By prioritizing the detection of the minority class through class weights, the model achieved a Sensitivity of 61.41%, identifying 646 non-survivors. The Negative Predictive Value (NPV) was 93.69%, highlighting the model's reliability in identifying patients at low risk of in-hospital death.

In order to achieve this level of sensitivity, the model accepts a higher rate of false positives (n = 2,515), resulting in a Positive Predictive Value (PPV) of 20.44%. In a critical care context, this trade-off is clinically acceptable, as over-monitoring a surviving patient is preferable to overlooking a patient at high risk of mortality. Finally, while overall accuracy was 69.57%, the Balanced Accuracy of 65.99% provides a more realistic measure of the effectiveness of the model across both outcome classes.

Table 3. Model Performance and classification statistics on the test set (n = 9,599).

Metric	Value	Interpretation
AUC-ROC	7.255	Overall discriminatory power
Sensitivity (Recall)	61.41%	Ability to correctly identify non-survivors
Specificity	70.57%	Ability to correctly identify survivors
Negative Predictive Value (NPV)	93.69%	Probability that a predicted survivor actually survives
Positive Predictive Value (PPV)	20.44%	Probability that a predicted non-survivor actually dies
Accuracy	69.57%	Overall percentage of correct predictions
95% Confidence Interval	(68.64% – 70.49%)	Precision range for the accuracy estimate
Balanced Accuracy	65.99%	Mean of sensitivity and specificity (class-adjusted)

Figure 2. Confusion matrix heatmap for in-hospital mortality prediction on the test set (n = 9,599).

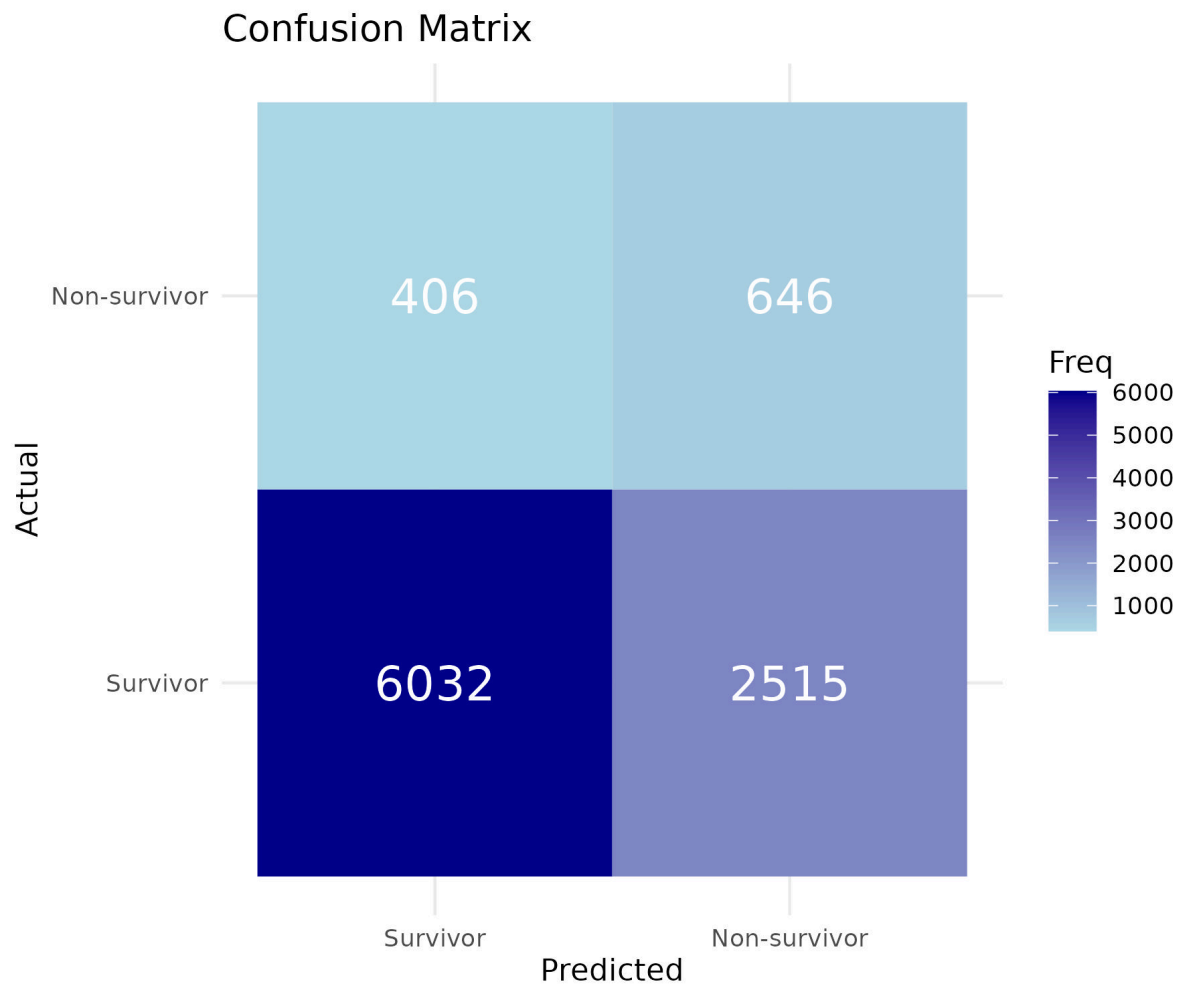
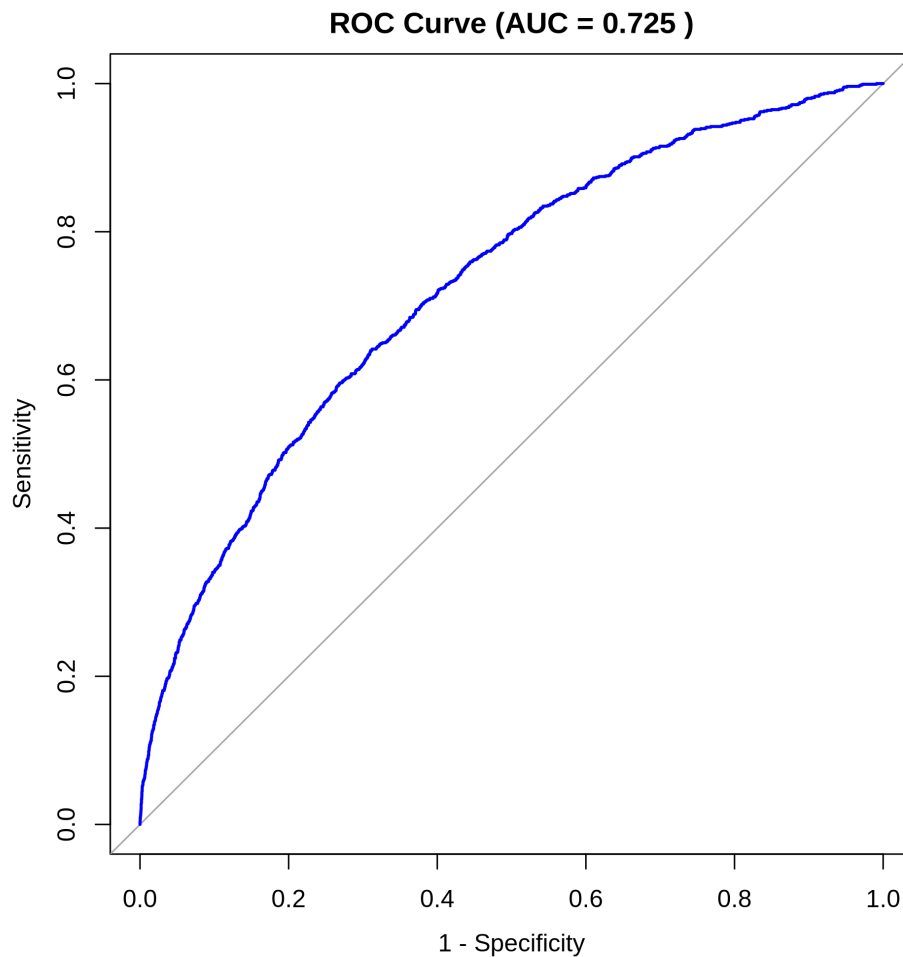


Figure 3. Receiver operating characteristic (ROC) curve for the multivariate logistic regression model predicting hospital mortality.



Note: The ROC curve illustrates the discriminatory performance of the model based on clinical variables collected during the first 24 hours of ICU admission. The area under the curve (AUC) was 0.714, indicating moderate discriminatory ability

Overall, the model achieved a level of performance comparable to that reported for early risk stratification models based on routinely collected ICU data, while maintaining interpretability and clinical transparency.

Discussion

In this retrospective analysis of a large ICU cohort derived from the MIMIC-III database, we developed an interpretable multivariable logistic regression model to identify early predictors of in-hospital mortality using routinely collected electronic health record data from the first 24 hours of ICU admission. The final model integrates demographic characteristics, physiological variables, and laboratory markers of early organ dysfunction and demonstrated moderate discriminatory performance, with an area under the receiver operating characteristic curve (AUC) of 0.7255 (Table 2). This level of performance is comparable to that reported by other early ICU risk stratification models based on limited initial data.

Age emerged as one of the strongest independent predictors of in-hospital mortality, with increasing age associated with progressively higher odds of death (Table 3). This finding is consistent with previous ICU studies and likely reflects reduced physiological reserve, a higher burden of comorbidities, and increased vulnerability to acute organ failure in older patients. Sex-related differences were also observed, with male sex associated with a modest reduction in adjusted mortality risk (Table 3). This association has been variably reported in the literature and may reflect differences in case mix, baseline health status, or ICU admission patterns rather than a direct biological effect.

Markers of early physiological stress were strongly associated with mortality. Higher heart rate and respiratory rate during the first 24 hours independently predicted an increased risk of death (Table 3), likely reflecting sustained sympathetic activation, respiratory distress, or compensatory responses to early metabolic derangements. In contrast, mean arterial pressure and oxygen saturation were not independently associated with mortality in the fully adjusted model, as their effects did not reach statistical significance (Table 3). Although these variables differed significantly between survivors and non-survivors in univariate analyses (Table 1), their prognostic value appears to be mediated through other clinical and laboratory factors captured in the multivariable model.

Among laboratory parameters, lactate and creatinine were the two strongest independent predictors of in-hospital mortality. Elevated lactate demonstrated the largest effect size, with an odds ratio of 1.459 per mmol/L increase (95% CI 1.42–1.50), reinforcing its role as a robust marker of tissue hypoperfusion and metabolic stress (Table 3). Similarly, creatinine showed a strong association with mortality, with an odds ratio of 1.305 (95% CI 1.27–1.34), highlighting the prognostic importance of early renal dysfunction and acute kidney injury in critically ill patients. Compared with earlier model iterations, the increased magnitude of these odds ratios suggests improved sensitivity of the updated model to markers of early organ failure.

Importantly, the application of a weighting strategy during model development improved the identification of patients who died during hospitalization. This is reflected in a sensitivity of 61.4% and a high negative predictive value of 93.7% (Table 2), indicating that the model performs particularly well in identifying patients at low risk of mortality. From a clinical perspective, this characteristic is relevant for early risk stratification, as it allows clinicians to confidently rule out high-risk patients and may support prioritization of monitoring and resource allocation in the ICU setting.

Overall, the updated model demonstrates improved discrimination while preserving transparency and interpretability. Although its predictive performance remains moderate, the use of a conventional

logistic regression framework allows clear attribution of risk to clinically meaningful variables. In clinical contexts where explainability and trust are essential, such as decision-support applications, this approach may offer practical advantages over more complex black-box machine learning models and should be viewed as a complementary tool rather than a definitive prognostic instrument.

Limitations of the study

This study has several limitations. First, the analysis was based on retrospective data from a single-center ICU database, which may limit generalizability to other institutions or healthcare systems. Second, EHR data are subject to missing values, measurement errors, and extreme outliers, as reflected by the high variability observed in some physiological variables. Third, the outcome assessed was in-hospital mortality rather than ICU-specific mortality, which may be influenced by post-ICU care and discharge practices. Finally, although the model demonstrated reasonable discrimination, external validation in independent cohorts would be required before clinical implementation.

Conclusion

In conclusion, routinely collected electronic health record data from the first 24 hours of ICU admission contain clinically meaningful information for early in-hospital mortality risk stratification. Our weighted logistic regression model identified, analyzed in 32,000 MIMIC-III admissions, identified age (OR 1.03), lactate (OR 1.459) and creatinine (OR 1.31) as the strongest predictors, surpassing traditional hemodynamic variables. With an AUC of 0.7255, the model was able to correctly identify 61.41% of deaths and offered 93.69% certainty in its survival predictions (NPV). These findings support the potential role of transparent, data-driven models as complementary tools in critical care decision-making and highlight the value of large EHR databases for clinical research.

Supplementary Material

Supplementary Notebook S1. SQL data extraction

Supplementary Notebook S2. Data preprocessing and modelling

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